2019 EXAMINATIONS

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.365 Advanced Networking

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Simple calculators permitted

Question 1

1.a Congestion avoidance and control is one of the functionalities offered by TCP.

i. Define the term *network congestion*. What is the impact of congestion on network performance if the end-points are not using a congestion control algorithm?

[3 marks]

ii. Bufferbloat is an important challenge for modern networks. Describe bufferbloat and discuss its impact on traditional congestion control algorithms.

[3 marks]

iii. Describe the TCP slow start phase. Discuss the importance of the slow start phase for TCP congestion control algorithms.

[2 marks]

1.b TCP offers reliable communication to network applications. Nonetheless, because the underlying packet delivery mechanism (IP) is best effort, TCP implementations use a range of packet loss detection mechanisms to detect and retransmit lost packets. Describe two packet loss detection mechanisms defined as part of the TCP standard.

[5 marks]

1.c Resource pre-allocation is a common approach used to multiplex multiple signals on a shared medium. Describe two resource pre-allocation multiplexing techniques used in wireless communications.

[4 marks]

1.d The TCP standards define a parameter to limit the number of packets that can be in-flight during the initial slow start phase of the TCP congestion window. This parameter is called *initial congestion window* and its value is suggested to be between 2-4 segments (2-4 KB) in RFC 3390.

In recent years, major Content Delivery Network (CDN) providers have been actively promoting an update of the TCP standard in order to increase the initial congestion window to 10 segments.

i. Why are CDN providers promoting this change? Discuss the impact that an increase in the initial congestion window would have on network performance. How is this change going to affect TCP fairness?

[4 marks]

ii. Applying your knowledge of Advanced Networking concepts, propose an *alternative strategy* that could be used to achieve comparable results.

[4 marks]

Total 25 marks

Question 2

2.a Device backplanes employ packet buffers to smooth traffic bursts and reduce packet loss. Describe each of the the different architectural approaches to packet buffering in a device backplane and discuss their trade-offs.

[6 marks]

2.b P4 is a dataplane programming language, aiming to replace OpenFlow as the next generation SDN de-facto standard. Identify three major design goals of the P4 language and discuss how the language and the compiler address them.

[6 marks]

2.c Packet scheduling between network queues is a key functionality for effective network resource control in a network device.

i. Describe *max-min fairness*. Discuss the main challenge to providing short-term max-min fairness guarantees in an Ethernet network.

[2 marks]

ii. Describe a scheduling algorithm used to approximate max-min fairness in an Ethernet forwarding device and discuss its drawbacks.

[3 marks]

Question 2 continues on next page…

Question 2 continued.

2.d A NAT and a WAN optimizer are two popular networking appliances, deployed in production enterprise networks.

A NAT is a networking appliance used to cope with address exhaustion in IPv4. A NAT device remaps an IP address space into another, by processing network and transport header fields on each packet. Typically, such devices are deployed on the edge of the network and allow devices on a local network to access Internet services using a limited number of public IP addresses. For each network flow, the device maps a local IP and port number to a public IP and port number and each incoming and outgoing packet is modified accordingly.

A WAN optimizer is a more complex networking appliance which aims to improve network resource utilization and responsiveness in a Wide Area Network (WAN). It monitors and transparently manipulates traffic by applying a wide range of latency and bandwidth reduction techniques, such as information compression and caching. A WAN optimizer can parse protocol information up to the application layer and uses application specific techniques to improve network utilization.

A network vendor aims to design a new generation of NAT and WAN optimizer appliances using SDN technologies. In order to implement these appliances, the vendor has access to two packet processing platforms. Platform A is a high-performance software platform supporting zero-copy packet processing and OS-level virtualization. Platform B uses an ASIC-based packet processor with P4 support.

The vendor wants you to comment on the suitability of each platform for the development of separate NAT and WAN appliances. What type of challenges do you expect the developers to face on each platform? Discuss the potential performance bottlenecks for each technology when implementing the aforementioned network appliances.

[8 marks]

Total 25 mark

Question 3

3.a Northbound and Southbound APIs are critical for Software Defined Networks (SDN). Explain the role of each of these APIs and how they support the overall ambition of enabling programmability in modern networks.

[5 marks]

3.b OpenFlow can operate in either *reactive* or *proactive* mode. Explain the behaviour of each mode, and the associated trade-offs between each approach.

[4 marks]

3.c Fog (or Edge) computing is a recent concept which unlocks new capability at the edge of networks. You have been asked, as a consultant, to consider how Network Function Virtualisation (NFV) may be employed at this network edge.

i. What are the potential benefits of using NFV at the edge of networks?

[3 marks]

ii. Suggest two network services or applications would benefit from having NFV at the edge.

[2 marks]

iii. What challenges exist when considering the deployment of NFV at the edge?

[3 marks]

3.d A new ISP is observing that a significant proportion of traffic going to their customers are adaptive-bitrate video streams supplied to the ISP via their transit provider.

i. Describe *three* different approaches they could adopt in order to reduce the financial burden associated with the supply of this traffic to their customers.

[6 marks]

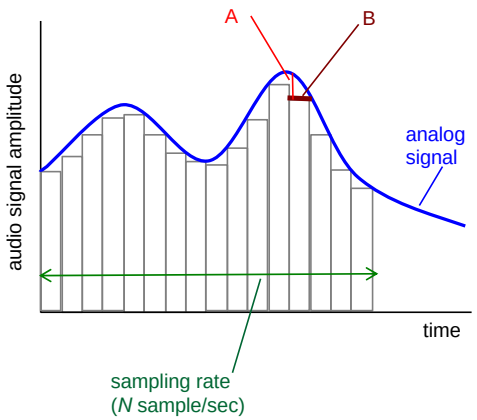
ii. Which of these solutions would you recommend if their priority was in maximising the video quality received by customers? Why?

[2 marks]

Total 25 marks

Question 4

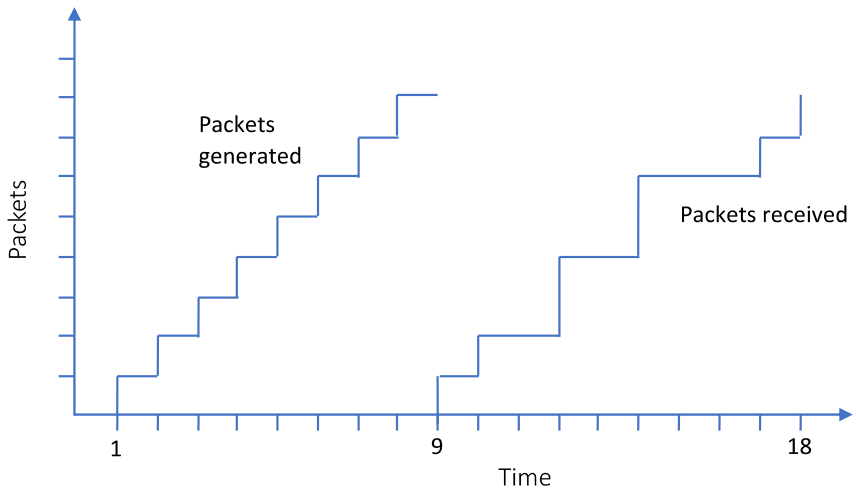
4.a The following diagram shows a representation of an analog audio signal being sampled at a constant rate.



Explain the overall process that is taking place in the Figure and identify specifically what the labels marked A and B represent.

[3 marks]

4.b A sender is transmitting packetized audio periodically, starting at time $t = 1$. The first packet arrives at the receiver at $t = 9$. The figure below shows the timing of the first eight packets transmitted by the sender and their subsequent arrival at the receiver.



i. If audio playout begins immediately as the first packet arrives at the receiver (t = 9), which of the first eight packets sent will not arrive in time for playout? **[1 mark]**

ii. What is the minimum playout delay at the receiver that results in all of the eight packets arriving in time for their playout? **[2 marks]**

Question 4 continues on next page…

Question 4 continued.

iii. Why would increasing the playout delay be problematic for certain types of applications? **[2 marks]**

iv. Discuss the potential issues of a playout strategy that uses a *fixed* playout delay. Describe an approach to resolving this. **[5 marks]**

4.c A company is experiencing drop-outs in their VoIP communications which they believe is due to them using a shared 1.5Mbps link for all their traffic. You have been asked by the company to investigate how they might make use of multiple classes of service.

i. Explain the four principles underpinning the mechanisms needed to provide quality of service (QoS). **[8 marks]**

ii. Would you recommend an approach using differentiated services or per-connection QoS guarantees? Explain the reason for your choice. **[4 marks]**

Total 25 marks

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